BMS COLLEGE OF ENGINEERING
Bengaluru -560019

Autonomous College, Affiliated to
Visvesvaraya Technology University, Belgaum

Alternative Assessment Tool (AAT) Report on

"Design and Verification of Combinational and Sequential Digital Modules using Verilog HDL"

Submitted in partial fulfilment of the requirements of the Alternative Assessment Tool (AAT) for the course

ADVANCED DIGITAL AND LOGIC DESIGN (ADLD)
Course Code: 23ECSPE1AD

Submitted By
MITHUN V - 1BM24EC111

Submitted To
Prof. AJAYKUMAR DEVARAPALLI

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
BMS College of Engineering
Bull Temple Road, Basavanagudi, Bengaluru-560019

Academic Year
2026 – 2027

TABLE OF CONTENT

SL.NO	TITLE	PAGE NO.
1	Arithmetic Logic Unit	1
2	Johnson Counter	4

1. ARITHMETIC LOGIC UNIT

TOOL USED: Cadence Design System.

CODE:

Design block

```
// 8 bit arithmetic logic unit

`timescale 1ns/1ps
module alu ( A, B, sel, out, cout);
  parameter n=8;
  input [n-1:0] A, B; //8 bit inputs
  input [2:0] sel;  // 3 bit selector
  output reg [n-1:0] out; //bit 8 output
  output reg cout; // 1 bit carry
  always @(*) begin
    out ={(n){1'b0}}; // default value of out to prevent latch synthesis
    cout = 1'b0; //default value of carry
    case (sel)
      3'b000: {cout,out} = A + B;  // addition (carry at cout)
      3'b001: {cout,out} = A - B;  // subtraction (borrow at cout)
      3'b010: out = A & B;  // bitwise AND
      3'b011: out = A | B;  // bitwise OR
      3'b100: out = A ^ B;  // bitwise XOR
      3'b101: out = A << 1; // logocal shift left by 1
      3'b110: out = A >> 1; //logical shift right by 1
      3'b111: out = (A == B); // equality comparison
      default : {cout,out} = {(n+1){1'b0}}; //default
    endcase
  end
endmodule
```

Test bench

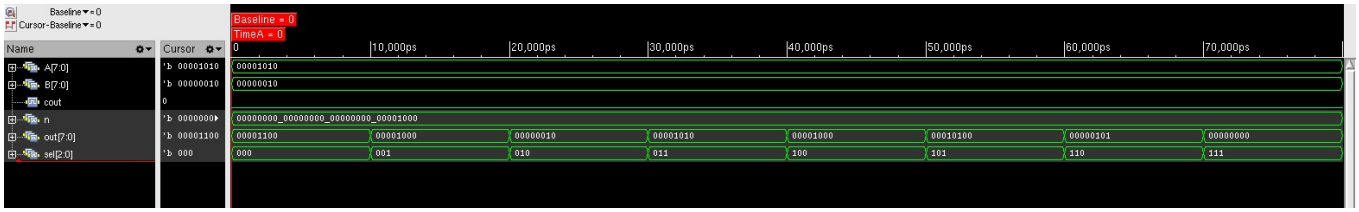
```
// test bench for 8 bit alu

`timescale 1ns / 1ps
module tb_alu;
parameter n=8;
reg [n-1:0]A,B;
reg [2:0] sel;
wire [n-1:0] out;
wire cout;
alu A1 (A,B,sel,out,cout);
initial begin
$dumpfile ("alu.vcd");
$dumpvars;
end
initial begin
$monitor ("Time = %0t | A = %d | B = %d | sel = %b | out = %d | cout = %d" ,   $time ,A ,B
,sel ,out ,cout);
A=8'h0A; B= 8'h02;
sel = 3'b000;#10
sel = 3'b001;#10
sel = 3'b010;#10
sel = 3'b011;#10
sel = 3'b100;#10
sel = 3'b101;#10
sel = 3'b110;#10
sel = 3'b111;#10
$finish;
end
endmodule
```

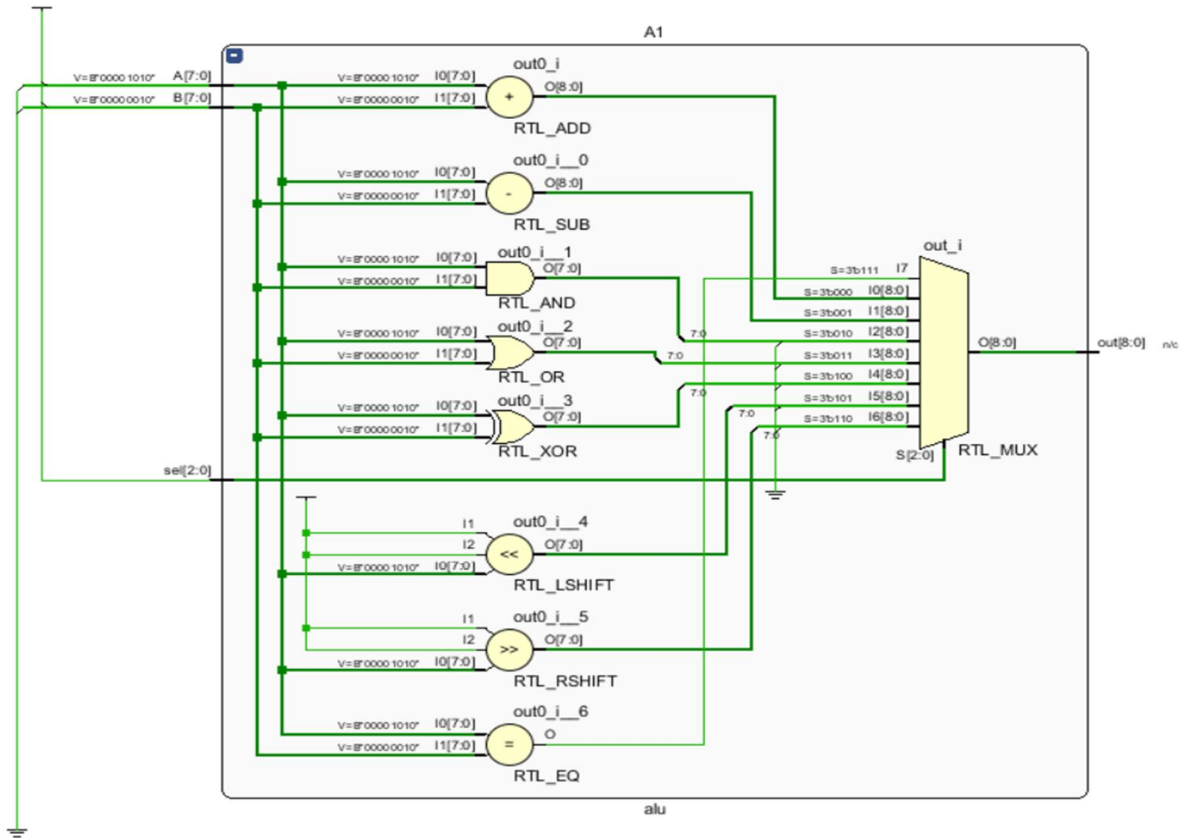
OUTPUT:

```
ncsim> source /opt/cadence/installs/INCISIVE152/tools/inca/files/ncsimrc
ncsim> run
Time = 0 | A =  10 | B =   2 | sel = 000 | out =  12 | cout = 0
Time = 10000 | A =  10 | B =   2 | sel = 001 | out =   8 | cout = 0
Time = 20000 | A =  10 | B =   2 | sel = 010 | out =   2 | cout = 0
Time = 30000 | A =  10 | B =   2 | sel = 011 | out =  10 | cout = 0
Time = 40000 | A =  10 | B =   2 | sel = 100 | out =   8 | cout = 0
Time = 50000 | A =  10 | B =   2 | sel = 101 | out =  20 | cout = 0
Time = 60000 | A =  10 | B =   2 | sel = 110 | out =   5 | cout = 0
Time = 70000 | A =  10 | B =   2 | sel = 111 | out =   0 | cout = 0
Simulation complete via $finish(1) at time 80 NS + 0
../Tb/tb_alu.v:29 $finish;
ncsim> exit
[student@ECE Work]$
```

WAVEFORM:



SCHEMATIC DIAGRAM:



2. JOHNSON COUNTER

TOOL USED: Cadence Design System.

CODE:

Design block

```
// 4 bit johnson counter

`timescale 1ns / 1ps
module johnson_counter (clk, reset, q);
input clk;
input reset;
output reg [3:0]q;
always @(posedge clk) begin
        if (reset) begin
                q <= 4'b0000;
        end
        else begin
                q = {q[2:0],~q[3]};
        end
end
endmodule
```

Test bench

```
//test bench for 4 bit johnson counter

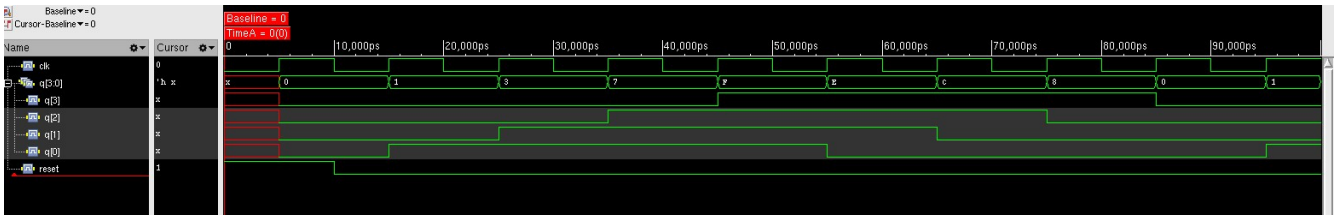
`timescale 1ns / 1ps
module tb_johnson_counter;
reg clk;
```

```
reg reset;
wire [3:0] q;
johnson_counter A1 (clk, reset, q);
always #5 clk = ~clk;
initial begin
$dumpfile ("johnson_counter.vcd");
$dumpvars;
end
initial begin
$monitor ("Time = %0t | Reset = %b | Q = %b ",
                        $time, reset, q);
clk = 0; reset = 1; #10
reset = 0; #90
$finish;
end
endmodule
```

OUTPUT:

```
ncsim> source /opt/cadence/installs/INCISIVE152/tools/inca/files/ncsimrc
ncsim> run
Time = 0 | Reset = 1 | Q = xxxx
Time = 5000 | Reset = 1 | Q = 0000
Time = 10000 | Reset = 0 | Q = 0000
Time = 15000 | Reset = 0 | Q = 0001
Time = 25000 | Reset = 0 | Q = 0011
Time = 35000 | Reset = 0 | Q = 0111
Time = 45000 | Reset = 0 | Q = 1111
Time = 55000 | Reset = 0 | Q = 1110
Time = 65000 | Reset = 0 | Q = 1100
Time = 75000 | Reset = 0 | Q = 1000
Time = 85000 | Reset = 0 | Q = 0000
Time = 95000 | Reset = 0 | Q = 0001
Simulation complete via $finish(1) at time 100 NS + 0
../TB/tb_johnson_counter.v:24 $finish;
ncsim> exit
```

WAVEFORM:



SCHEMATIC DIAGRAM:

